Chapter 3: Dimensionality Reduction

Outline

3.1 Introduction

- Unsupervised learning (UL) is a machine learning area in which algorithms extract features and identify commonalities in a dataset that has not been labeled or classified.
- Dimensionality reduction (DR) and clustering are two key aspects of UL.

3.2 Data Mining

- One of the main uses of UL is pattern recognition.
- The process of finding patterns on unlabeled data is also known as data mining.
- In process applications, data minings enbles a better understanding of the system by identifying its differen modes of operation.

Dimensionality reduction

3.3 Dimensionality Reduction - Methods Overview

- DR is the process of reducing the number of variables or attributes that are taken into consideration.
- The goal is to find a lower dimensional representation of the original data tha still preserves most of the information.
- Broadly speaking, there are two may groups of DR:
  - Matrix factorization
  - Neighbor graphs or manifold learning

3.3.1 Matrix Factorization - PCA

- Principal component analysis (PCA) is a linear distance preservation technique that determines the set of orthogonal vectors that optimally capture the variability of the data.
- Given a set of n observations and m sensor measurements stored in $\mathbf{X} \in \mathbb{R}^{n \times m}$, its sample covariance matrix $\mathbf{S}$ is:

$$\mathbf{S} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^T \tag{1}$$

- By finding the eigenvalues $\boldsymbol{\Lambda}$ of $\mathbf{S}$, the lower dimensional representation is

$$\mathbf{T} = \mathbf{X}\mathbf{P} \tag{2}$$

- In practice, only a few of the vectors in $\mathbf{P}$, those that correspond to the largest singular values, are used for the projection.

3.3.2 Matrix Factorization - Improvements

- Independent component analysis (ICA)
  - Find components that are statistically independent from each other
- Kernel principal component analysis (kPCA)
  - Projects data into a higher-dimensional space using a kernel function to make it linearly separable
  - Performs PCA in the transformed space to capture nonlinear relationships in the original data

3.3.3 Neighbor Graphs - t-SNE

- t-distributed stochastic neighbor embedding (t-SNE) (Van der Maaten et al. 2008) produces a lower dimensional projection by finding the pairwise distances between points in the original space and then finding the pairwise distances in the lower dimensional space that minimizes the Kullback-Leibler divergences between the two.

$$\mathrm{KL}(P \parallel Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \tag{3}$$

3.3.4 Neighbor Graphs - Improvements

- Uniform manifold approximation and projection (UMAP) (McInnes et al. 2018) creates a neighbor graph of the high dimensional space (W^H) and a neighbor graph in the lower dimensional space (W^L) then, the final graph in determined by minimizing the cross entropy between the two graphs:

$$\mathcal{L} = \sum_{i \neq j} W_{ij}^H \log \frac{W_{ij}^H}{W_{ij}^L} + (1 - W^H)_{ij} \log \frac{1 - W_{ij}^H}{1 - W_{ij}^L} \qquad (4)$$

3.3.5 Self-organizing Maps

- Self-organizing maps (SOMs) aim to preserve the topological structure of the data by mapping high-dimensional data onto a lower-dimensional grid while maintaining the relative distances between points.
- The principle of SOMs is to iteratively adjust the weights of neurons in a grid to match the input data, such that similar data points are mapped to nearby neurons, forming clusters and preserving the data's topology.
- SOMs can produce different visualizations, such as component planes, distance matrices, and projections.

3.4 Global and Local Structure

- One of the goals of DR is to preserve the structure of the original high-dimensional space as it is projected into a lower dimensional representation.
- In general, DR algorithm focus on the global structure of the data (PCA) or local structures (t-SNE) to achieve this.

3.4 Global and Local Structure (continued)

- Pairwise Controlled Manifold Approximation Projection (PaCMAP) (Wang et al. 2021) is a DR technique that balances the preservation of both local and global structures in the data.
- PaCMAP optimizes a cost function that emphasizes three types of point pairs: neighbors, mid-near pairs, and further pairs, to achieve a balance between local and global structure preservation.
- It is particularly effective for visualizing high-dimensional data while maintaining meaningful relationships between data points.

3.5 Conclusions

- Dimensionality reduction is a key aspect of unsupervised learning, enabling better understanding and visualization of high-dimensional data.
- Techniques like PCA, ICA, t-SNE, UMAP, and PaCMAP offer various approaches to reduce dimensions while preserving data structure.
- The choice of method depends on the trade-offs between preserving global and local structures and the specific application requirements.
- Dimensionality reduction facilitates pattern recognition, clustering, and insights into system behavior in process applications.